Can random transformations neutralise a universal perturbation?

What the literature says, and what we can expect for our project. (English translation of defenses_aleatoires.pdf, written before our own experiments.)

In short

- The idea was tried as early as 2017–2018: randomly resize, shift, compress or add noise to the image before feeding it to the network (Xie et al.; Guo et al.). Against attacks that ignore the defense, the results are spectacular (from 0% to 97–99% accuracy).
- **All these defenses were broken** by attacks that know the transformation (Athalye, Carlini and Wagner, 2018). The key tool is **EOT**: optimise the perturbation on average over all the random transformations. Result: 0% accuracy.
- A universal perturbation lends itself particularly well to EOT: the **Adversarial Patch** (Brown et al., 2017) is universal and resists random rotations, rescaling and positions.
- The only proven random defense, **randomized smoothing** (Cohen et al., 2019), only guarantees robustness for ℓ_2 perturbations of norm ≤ 0.5 to 3, whereas our ResNet-50 perturbation has $\|v\|_2 = 10.46$.
- **For our project**: randomness will probably weaken our current perturbation (computed without taking it into account), but a perturbation recomputed with EOT should recover its effectiveness. It is an instructive experiment, not a defense.

Note added after the experiments. Both predictions were confirmed on ImageNet with Gaussian noise $\sigma = 20/255$: the non-adaptive perturbation of ResNet-50 drops to 14% fooling, and a perturbation recomputed with EOT recovers 45–50% at the same budget and 86% at $\xi = 16/255$. See the project overview.

1. Essential vocabulary

Adversarial perturbation. A small vector r added to an image x such that the classifier changes its mind: the class predicted for x + r differs from the one predicted for x. It is usually computed for one image. A **universal** perturbation v is a single vector that works for most images (our topic).

Random pre-processing defense. The network f is left untouched; the input is transformed by a transformation t_θ drawn at random (θ = size, position, noise…) at every prediction: the defended classifier is $f(t_\theta(x))$. The hope: the transformation "breaks" the fine structure of the perturbation, and the attacker does not know which transformation will be drawn.

Threat models — what the attacker knows. This is the point that explains everything else:

Model	The attacker knows…	Comment
Black box	nothing about the model; it attacks another model and hopes the attack transfers	the weakest
Gray box	the network and its weights, but not the defense	setting of Xie and Guo
White box	everything, including the defense and the distribution of the randomness (but not the draw)	
Adaptive attack	white box + an attack designed specifically against this defense	standard required since 2018

Kerckhoffs's principle, borrowed from cryptography, applies: a defense must resist an attacker who knows it. A defense that only holds because the attacker ignores it is "security through obscurity".

2. Xie et al. (ICLR 2018): random resizing and padding

Paper: Mitigating Adversarial Effects Through Randomization (arxiv.org/abs/1711.01991). This is exactly the idea of "putting some randomness into every image".

Method

- **Random resizing**: the 299×299 image is resized to r×r, with an integer r drawn in [299, 331).
- **Random padding**: the resized image is padded with zeros up to 331×331, at a random position.
- That makes 12,528 possible transformations. Both layers are added at prediction time to already trained networks, without retraining.
- Tested on Inception-v3, ResNet-v2-101, Inception-ResNet-v2 and an adversarially trained version, against FGSM, DeepFool and C&W.

Three attack scenarios, from weakest to strongest

- **Vanilla**: the attacker attacks the network alone, without knowing the defense.
- **Single-pattern**: it attacks the network + one fixed transformation (centred image, 16-pixel padding, no resizing).
- **Ensemble-pattern**: it attacks an ensemble of 21 predefined transformations.

Results (top-1 accuracy on attacked images, without defense → with defense)

Model / attack	Vanilla	Ensemble-pattern
Inception-v3, DeepFool	0% → 98.3%	0.6% → 81.3%
Inception-v3, C&W	0% → 96.9%	0.6% → 62.9%
ResNet-v2-101, C&W	0% → 97.1%	1.0% → 74.3%
Inception-v3, FGSM ε = 10	33.0% → 52.4%	—

The accuracy on clean images only drops by 0.7 to 2.7 points. At the NIPS 2017 competition, the defense combined with an adversarially trained model obtained a normalised score of 0.924, against 0.773 for adversarial training alone.

Explanation and limitations (according to the authors)

- **Why it works**: iterative attacks (DeepFool, C&W) "stick" to the exact network they attack; their perturbations are very fine and a one-pixel shift is enough to break them. In their diagnosis, a one-pixel padding raises the accuracy under C&W from 0.8% to 70.2%, and a one-pixel resizing from 2.6% to 97.8%.
- **Acknowledged limitations**: less effective against FGSM with a large ε (coarser perturbation); as soon as the attacker averages over several transformations (ensemble-pattern), the accuracy drops clearly (62.9% under C&W); it is impossible to attack the 12,528 transformations at once, so the strongest attack was not tested.

Takeaway. The last limitation is the most important: the defense was evaluated against attackers who only averaged over 21 transformations. Section 4 shows what happens when the attacker averages properly.

3. Guo et al. (ICLR 2018): image transformations

Paper: Countering Adversarial Images using Input Transformations (arxiv.org/abs/1711.00117). Same idea, with other transformations, on a ResNet-50:

Transformation	Random?	Principle
Crop + rescale	yes	average of the predictions over 30 random 90×90 crops, rescaled to 224×224

Transformation	Random?	Principle
Bit-depth reduction	no	each pixel coded on 3 bits instead of 8
JPEG compression	no	quality 75
Total variance minimisation (TVM)	yes	each pixel is kept with probability 0.5, then a “smooth” image is reconstructed
Image quilting	yes	each 5×5 patch is replaced by a patch of clean images drawn among its nearest neighbours

Results (gray box, normalised ℓ_2 distance 0.06): under DeepFool, 44.9% accuracy with cropping, 44.7% with TVM, 34.5% with quilting, against 1.8% for an adversarially trained model. Combining cropping + TVM + quilting reaches 51.5%. The abstract states that the best defense eliminates 60% of strong gray-box attacks and 90% of black-box attacks.

Why they thought it would hold: (1) TVM and quilting are not differentiable, so no gradient can be computed through them; (2) the randomness forces the attacker to find a perturbation that works “for the whole distribution of possible images”. Point (2) is exactly what a universal perturbation does… which is not good news for the defense.

4. Athalye, Carlini and Wagner (ICML 2018): everything is broken

Paper: Obfuscated Gradients Give a False Sense of Security (arxiv.org/abs/1802.00420). The reference paper to understand why the previous defenses looked solid.

The diagnosis: “obfuscated” gradients

Attacks (DeepFool, PGD, C&W) follow the gradient of the network. Many defenses do not make the network more robust: they only make the gradient useless. Three forms:

- **Shattered gradients**: a non-differentiable (JPEG, quilting) or numerically unstable operation;
- **Stochastic gradients**: the gradient depends on the random draw, and a single draw gives a bad direction (the case of the idea we started from);
- **Exploding or vanishing gradients**: very deep computations, for instance several passes through a network.

The two weapons

EOT (Expectation Over Transformation) — against randomness. Instead of attacking $f(t_\theta(x))$ for one draw θ, the attacker attacks the average:

$$\text{maximise } E_\theta[\ \text{loss}(\ f(\ t_\theta(x + r)\ )\ )\ ] \text{ with } \nabla E_\theta[\ \dots\ ] = E_\theta[\ \nabla \dots\ ]$$

In practice, at each step, several transformations are drawn, the gradient is computed for each, and they are averaged. The resulting perturbation works on average over all transformations, hence whatever the draw of the defense.

BPDA (Backward Pass Differentiable Approximation) — against non-differentiability. The true transformation is kept in the forward pass, and replaced in the backward pass by a differentiable approximation, often simply the identity (one acts “as if” JPEG changed nothing).

Results

Defense	Attack	Result
Xie et al. (random)	EOT over the distribution of transformations, $\varepsilon = 0.031$ in ℓ_∞ ($\approx 8/255$)	100% success of the targeted attack ($\approx$ 0% accuracy), even when requiring the image to be fooled 10 times out of 10
Guo et al. (transformations)	EOT for cropping, BPDA for JPEG and bit depth, EOT + BPDA for TVM and quilting	100% success, 0% accuracy for the best defense

Overall, of 9 defenses published at ICLR 2018, 7 relied on obfuscated gradients; 6 are fully circumvented and 1 partially.

Their recommendations for evaluating a defense

- Assume that the attacker knows the architecture, the training and the defense; only the random draw stays secret.
- Evaluate against **adaptive attacks**, designed after the defense and against it.
- Do not artificially limit the computation of the attacker.
- Warning signs: a one-step attack better than an iterative one; black box better than white box; an attack that does not reach 100% even without a size limit.

5. Brown et al. (2017): a universal attack that resists randomness

Paper: Adversarial Patch (arxiv.org/abs/1712.09665). It is not a defense, but it is the most direct evidence for our question.

A sticker (“patch”) that one places anywhere in a scene so that the network predicts a chosen class, for instance “toaster”. It is computed by:

$$\hat{p} = \text{argmax}_p \, E_{x,\,t,\,l}[\, \log P(\, \hat{y} \mid A(p, x, l, t)\,)\,]$$

where the average is over random images x (universality), random positions l and random transformations t (translation, rescaling, rotation): it is a variant of EOT. Printed on a colour printer and placed on a table next to a banana, it turns VGG16 from “banana” (97%) into “toaster” (99%).

Why this is decisive for our idea. The objective of the patch is exactly “universal + robust to random transformations”. If it can be reached for a sticker that undergoes rotations, rescaling and displacements, it can be reached for a universal perturbation subjected to a random resizing or shift: it is enough to add the average over the transformations to the algorithm. The authors explain that the patch works because it is more “salient” than real objects: the network, trained to find the main object, focuses on it. This is the idea of the dominant class again (our “spider web”).

6. Gao et al. (NeurIPS 2022): fundamental limits

Paper: On the Limitations of Stochastic Pre-processing Defenses (arxiv.org/abs/2206.09491). Four years later, a general analysis of these defenses.

Finding 1 — most of them do not have enough randomness. The authors redo the attacks without EOT, with a simple standard PGD attack. Result: against Xie et al., 200 iterations of PGD without EOT are enough for 100% success; against Guo et al., 99.0%. In other words, the randomness of these defenses barely hindered the attacker. Same case study with a random rotation between −90° and +90°: 100% success.

Finding 2 — an unavoidable trade-off between robustness and invariance. To put a lot of randomness without destroying the accuracy on clean images, the model must be retrained to become insensitive to the transformation. But the more insensitive it is, the more the transformation lets the perturbation through, and the less it protects. The authors show this trade-off theoretically and experimentally: these defenses “protect” by making the model more variable, and must take this variability back to remain useful.

Conclusion of the authors: these defenses bring no intrinsic robustness; other ways of using randomness must be found.

7. Cohen, Rosenfeld and Kolter (ICML 2019): randomness with a guarantee

Paper: Certified Adversarial Robustness via Randomized Smoothing (arxiv.org/abs/1902.02918). The rigorous version of “adding noise”.

The classifier f is replaced by a smoothed classifier g that votes over noisy copies of the image:

$$g(x) = \text{argmax}_c \; P(f(x + \varepsilon) = c), \; \varepsilon \sim N(0, \sigma^2 I)$$

If class A wins the vote with probability at least p_A and every other class at most p_B, then no perturbation of ℓ_2 norm smaller than

$$R = (\sigma / 2) \cdot (\Phi^{-1}(p_A) - \Phi^{-1}(p_B))$$

can change the prediction (Φ^{-1}: inverse of the Gaussian cumulative distribution function). It is a proof, valid against any attacker, including adaptive ones. On ImageNet:

Guaranteed ℓ_2 radius	σ used	Certified accuracy	Accuracy on clean images
0.5	0.25	49%	67%
1.0	0.50	37%	57%
2.0	0.50	19%	57%
3.0	1.00	12%	44%

Limitations: the price in accuracy is high (67% at best, against ~76% for a standard ResNet-50); the model must be trained with Gaussian noise; prediction requires thousands of draws (100,000 to certify); the guarantee only covers the ℓ_2 norm.

Comparison with our perturbation. Our ResNet-50 perturbation has $\|v\|_\infty = 10/255$ but $\|v\|_2 = 10.46$ (images in [0, 1]). The guaranteed radii are 0.5 to 3: **three to twenty times too small**. Randomized smoothing therefore gives no guarantee against a universal perturbation of this size. This is the fundamental problem: a perturbation that is imperceptible pixel by pixel (ℓ_∞) is large in ℓ_2 norm in dimension 150,528.

8. And for universal perturbations in particular?

In the original paper (Moosavi-Dezfooli et al., 2017), the authors test a defense by retraining: VGG-F is retrained for 5 epochs on images to which, one time out of two, a universal perturbation drawn from 10 precomputed ones is added. The fooling rate of a perturbation recomputed on the retrained model goes from 93.7% to 76.2%, and repeating the procedure does not improve things. Conclusion of the authors: a modest improvement, not a protection.

Dedicated defenses. Borkar et al. (CVPR 2020) do not transform the image: they regenerate some features inside the network, those most sensitive to universal perturbations, and report that a defense trained against one universal attack also protects against others. I could only read the abstract of this paper: to be checked before relying on its numbers.

I did not find a study dedicated specifically to random transformations against universal perturbations (beyond a thesis of which I could only see an excerpt). This is a small gap that our project can fill at its scale.

9. Summary

Approach	What was claimed	Against an attacker who knows the defense
Random resizing + padding (Xie)	DeepFool, C&W: 0% → 97–98%	0% (EOT); even PGD without EOT reaches 100% success
Transformations (Guo)	up to 51.5% under DeepFool	0% (EOT + BPDA); PGD without EOT: 99% success
Very random transformations	robustness increasing with randomness	robustness / invariance trade-off (Gao)
Randomized smoothing (Cohen)	proven guarantee	holds, but only for $\|r\|_2 \leq 0.5$–3, and costs ~10–30 points of accuracy
Retraining on universal perturbations	—	93.7% → 76.2% fooling: modest

10. What would work, and what would not, in our project

What would probably work

- **Weakening our current perturbations.** They were computed without any transformation, and DeepFool only crosses the boundaries by 2%. We have already observed their fragility: on Imagenette, simply clipping to [0, 1] took the rate on X from 85.6% to 51.2%. Xie et al. show that a one-pixel shift often suffices to break a DeepFool perturbation. A random resizing or shift should move the “spider web” with respect to the pixel grid and clearly lower the fooling rate.
- **With one reservation**: the ResNet-50 perturbation on ImageNet is “coarser” (a texture repeated over the whole image, $\|v\|_2 = 10.46$), it could resist better than a single-image DeepFool perturbation. This has to be measured.

What would not work

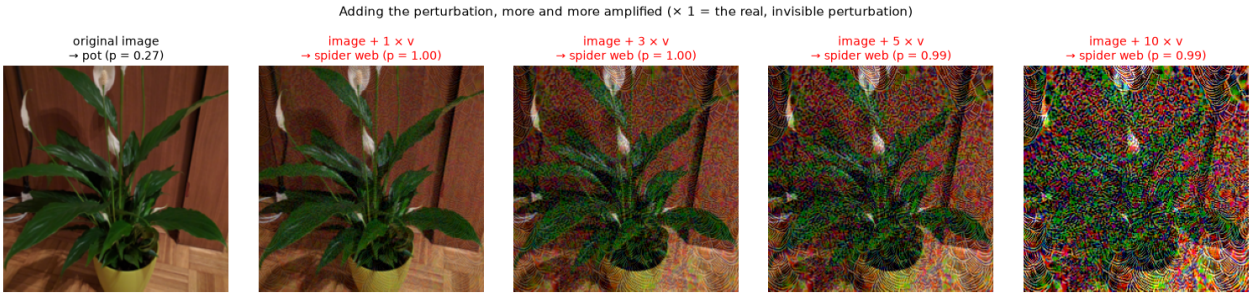
- **As a real defense.** It is enough to recompute the universal perturbation by averaging over the random transformations (EOT). It is particularly easy here: the universal algorithm already averages over thousands of images; adding a transformation draw for each image changes almost nothing. The Adversarial Patch shows that the result stays effective.
- **With much more randomness**: the accuracy on clean images drops, and retraining the model to recover it reduces the protection (Gao).
- **With “certified” Gaussian noise**: the certified radii are far too small compared with $\|v\|_2 = 10.46$.

The experiment we could run

It will not give a new defense, but it will reproduce, on our own perturbations, the lesson of the literature:

Step	What we do	What we measure	Expected
1. Naive defense	evaluate our perturbations after random resizing + padding, Gaussian noise, blur, JPEG	fooling rate; accuracy on clean images	clearly lower fooling rate, accuracy almost intact
2. EOT counter-attack	recompute the universal perturbation applying a random transformation to each image of the algorithm	fooling rate with the defense	back to a high fooling rate
3. Price of the defense	increase the intensity of the randomness	clean accuracy vs EOT fooling rate	a lot of accuracy must be sacrificed

What has to be written in the code: a random-transformation option in `evaluate.py` (step 1) and an EOT option in `universal_pert.py` (step 2). Step 1 can run on a PC with Imagenette; step 2 needs a GPU.



Our ResNet-50 universal perturbation, added to an image and amplified (× 1 = the real, invisible perturbation). It is this fine structure (the “spider web”) that a random transformation could shift or blur.